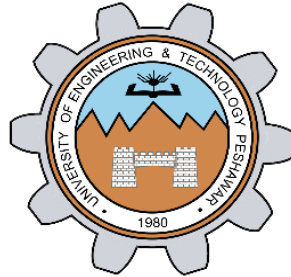


CHASHM AI

(Camera-based Hybrid Assistive Smart Headset Machine using AI)



SESSION 2022-2026

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April 2026

AUTHOR'S DECLARATION

I hereby declare that I am the sole author of this thesis. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners. It is further declared, that I have fulfilled all the requirements in line with the Quality Assurance guidelines of the Higher Education Commission.

I understand that my thesis may be made electronically available to the public.

Acknowledgements

It is our great pleasure to express our sincere thanks and heartfelt appreciation to our supervisor Dr. Madiha Sher, for all the support they have provided us throughout this project. Their expertise and suggestions contributed immensely towards our efforts. Further, we thank the faculty and staff members of the department of computer systems engineering at UET Peshawar for providing us the necessary laboratory facilities. Sincere thanks go to our fellow students and friends who have helped us in testing our prototypes and evaluating them. Lastly, we thank our families for their understanding and cooperation during this study period.

Abstract

The research introduces **CHASHM AI** as a new assistive headset system which enables visually impaired people to navigate better and retrieve information more effectively. The system operates through an ESP32-CAM module which runs on smart glasses to transmit real-time video feed to a Flutter-based mobile application. The mobile application uses artificial intelligence (AI) models to perform two essential functions which include detecting obstacles in front of users and recognizing printed text through **optical character recognition (OCR)**.

The system converts detected obstacles and recognized text into audio feedback which Bluetooth earbuds deliver for hands-free real-time guidance. The system uses deep learning algorithms running on devices for YOLO-based object detection and Tesseract or ML Kit OCR engines to understand the surroundings. The system demonstrated successful performance in its first prototype tests by detecting nearby obstacles including walls and vehicles and people with precision and by correctly reading aloud standard text materials more than 90% of the time when lighting conditions were favorable. The design of this compact low-cost device focuses on portability and user-friendly operation because it targets users who need budget-friendly assistive technology. The research shows that mobile devices can run camera vision systems with AI to create better self-navigation capabilities for people who are blind. The research uses Assistive Technology as its main subject while studying Visually Impaired people who use Smart Glasses for obstacle detection and Optical Character Recognition

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
OCR	Optical Character Recognition
TTS	Text-to-Speech
VI	Visually Impaired
ESP32	Espressif Systems 32-bit microcontroller
BT	Bluetooth
BLE	Bluetooth Low Energy
UI	User Interface
HEC	Higher Education Commission
IoT	Internet of Things

Table 1 Abbreviations Used

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CHAPTER 1

1 INTRODUCTION

1.1 Overview of Computer Vision in Assistive Technology

The pace of technological advancement has significantly increased in recent decades, and the development of new types of equipment is rapidly growing. For example, technology used by visually impaired people has drastically transformed due to the introduction of Computer Vision and Artificial Intelligence. With the help of such technologies, systems can detect any object, recognize texts, and deliver information about the surrounding environment. Therefore, people who have vision impairment can perform numerous routine tasks independently.

While there is always visual information available in the environment, its transformation into useful data is often difficult. Firstly, a visually impaired individual would need the system to operate almost instantly. Otherwise, it would become useless. Additionally, any confusion or unclear feedback will prevent visually impaired people from achieving their goal. However, at the same time, systems should be lightweight and comfortable to wear continuously.

Daily tasks such as walking around unfamiliar places or detecting obstacles might be challenging for visually impaired individuals. They might use traditional aids like white canes, but these means cannot provide enough information about what is located ahead. It makes navigation a difficult process in some cases.

Existing systems are ineffective outside the experimental setting in many cases. Either the process of transforming the obtained data takes too much time, or the audio feedback is not easy to comprehend. In both cases, any delay prevents movement, and the visually impaired person would need additional efforts to continue his or her actions.

On the other hand, there are numerous problems when it comes to developing such a device for visually impaired people. First of all, the device is expected to be light but efficient enough to execute its functions. Secondly, developers may struggle with

deciding which function to give priority to – to detect obstacles or assist with navigating or recognize text. Lastly, one should take into account issues connected to the use of the auditory system. Recent developments in the areas of computer vision, edge computing, and natural language processing improved assistive technologies greatly. Nowadays, the models based on deep learning algorithms manage to identify objects and texts efficiently. However, the majority of current applications can be characterized by limited functionality. Some can help users navigate around, some are able to recognize text, but there is no universal app that performs all these functions at once.

1.2 Rationale for CHASHM AI Development

The primary reason for creating CHASHM AI was based on the observation of limitations of current assistive technologies. Most visually impaired people need to use several tools separately for each task. While some might be helpful in obstacle detection, others are useful for reading printed texts. Having and controlling all those devices might be troublesome.

CHASHM AI is going to unite all those functions in a single device that can be worn easily and provide assistance with navigation and text reading at the same time. It will be able to recognize objects in its surroundings and announce printed texts aloud without the need for additional equipment or help from another person.

As it was said earlier, the main components of the proposed solution include a wearable device, a smartphone, and the algorithm implemented in the latter. The headband camera captures live video streams and sends them to the user's smartphone via Bluetooth. AI technology analyzes them, and the smartphone sends audio information about the recognized images.

The system is expected to work effectively and efficiently. Audio notifications can warn the user about potential obstacles in the path and also read aloud any printed text available in documents, books, or signboards. In addition to this, the proposed application can provide more convenient navigation experience.

While most current technologies replace traditional navigation aids, CHASHM AI is supposed to work together with those tools. In combination with clear audio feedback,

it can make life easier for the visually impaired people. It will help to navigate unknown places without additional stress due to lack of understanding of the situation.

Also, the proposed system will allow users to read on-demand any printed text. This function will be helpful for studying notes, official documents, and other written materials that cannot be used conveniently without any additional assistance.

As far as technical details are concerned, there is no necessity to process the collected data on the headset itself. Video frames should be transmitted to the smartphone using Bluetooth. Intensive processing and recognition tasks can be performed by smartphones since they already have enough computing power.

1.3 Problem Statement

The visually impaired find it challenging to understand the environment and perceive any printed information in real-time. In the process of walking and navigating, people must perceive any obstacles in order to move around safely. On top of that, people are supposed to recognize texts and other symbols written in any document or book. Nevertheless, the lack of sight creates difficulties as one cannot perform all the above operations in a fast manner when there are a lot of visual stimuli around. Such traditional instruments as white canes or handheld magnifiers help in some extent, yet these devices can hardly provide any additional information about the environment and are unable to read printed text immediately.

On the contrary, the number of visual stimuli in our lives continuously grows. These stimuli include traffic lights, labels, signs in shops, written documents, books, screens, etc. All these things create sensory overload for visually impaired individuals who simply cannot get information from the environment because the latter is visual in nature and cannot be turned into something else. Consequently, it means that visually impaired persons miss a lot of valuable information or struggle when making safe decisions and movements.

Advanced computer vision systems can potentially provide visually impaired people with needed assistance. At the same time, these systems require powerful processors that use a lot of energy. Such characteristics make it almost impossible to integrate a CV system into a wearable device such as smart glasses since wearing them for a long

time would create discomfort and overheating. Therefore, it is crucial to perform high-performance processing in another device which is easy for a user to carry, i.e., a smartphone.

As of now, assistive devices suffer from the following significant limitations:

- Obstacles detection vs. text recognition inability;
- The ability to recognize text while the inability to provide navigation assistance during movement;
- Delay in analysis that exceeds 100 ms;
- Too light devices with poor performance capabilities or powerful devices that are too bulky to wear;
- Lack of navigation-text reading combinations and integration into a single smooth and hands-free experience.

Thus, visually impaired persons still lack comprehensive solutions that enable confident and safe navigation as well as recognition and reading of text without any effort.

These facts highlight the necessity of developing a new type of solution:

- The wearable and lightweight headset;
- A rapid capture of visual data with

1.4 Research Objectives

Some basic goals also exist within the CHASHM AI project, which have been developed in order to guide the development of the project itself.

1.4.1 Primary Goals

1.4.1.1 Goal_1:

The hybrid system design shall feature the ESP32-S3 CAM as a device to process the video and transmit the video to the mobile phone application for AIT purposes. Such a design will ensure lightweight performance of the headset.

1.4.1.2 Goal_2:

Create the fast video streaming protocol based on RTP/UDP, in order to ensure that the total delay remains at below 100 ms. Such a solution is critical in relation to the safety during the navigation.

1.4.1.3 Goal_3:

Integrate visual machine learning algorithms, for instance, smaller-sized variants of YOLO and Mobile Net SSD into the mobile application. With the aid of such integration, the application will be capable of rapid identification of the object or the scene and thus the user will be able to avoid any kind of obstacle.

1.4.1.4 Goal_4:

Create a text recognition and reading module, whereby the user will be able to scan the text from any piece of paper and listen to its playback via an audio output in the earbuds. For this purpose, an OCR engine will be used on the mobile phone.

1.4.1.5 Goal_5:

Deliver good quality audio alerts and reading by equipping the earbuds with high-quality components

1.4.2 Secondary Goals

1.4.2.1 Goal 6:

We hope to create an ergonomic hardware housing for our device to make its prolonged use comfortable and safe for the wearer. In order to accomplish such an objective, we need to focus on designing a robust physical casing that uses light yet efficient configurations in order not to drain the battery pack while using the ESP32-S3 chip continuously.

1.4.2.2 Goal 7:

In order to ensure the viability of our product when used by blind individuals, we plan to test it in the field by analyzing precise milliseconds needed for processing the stream of information and gathering feedback from actual users regarding their confidence and comfort when navigating with it.

1.4.2.3 Goal 8:

The application we develop should offer a straightforward and clean user interface which would be convenient for people who have difficulty using visual interfaces. Such a UI must allow the user to change between modes and adjust the settings with simple and clear gestures as fast as possible.

1.4.2.4 Goal 9:

It is important for our software architecture to include an error-handling module that will track exceptions, such as connection drops, frame pauses, and increased processing latency, and inform the user about this issue in real-time through voice commands.

1.4.2.5 Goal 10:

As a concluding step, we will write detailed technical documentation regarding the development of our system including circuit wiring schematics, UDP and RTP protocol implementation, and mobile application optimization. These logs are aimed to provide other researchers with a detailed roadmap of how to develop a reliable and low-latency assistive technology solution.

1.5 Project Scope and Deliverables

The CHASHM AI project seeks to design and develop an intelligent headset and an advanced mobile app with all possible functionalities for visually impaired persons to help them navigate, perceive their surroundings, and read written texts.

The CHASHM AI scope includes developing a functional and cost-effective, as well as real-time and intelligent system, which uses Artificial Intelligence, computer vision, and wireless communications technologies.

The following areas are to be covered by the project:

1.5.1 Functional Scope

The functional scope of CHASHM AI involves determining functions performed by the system to provide the benefit to the user.

- The ESP32-S3 camera installed in the headset will capture video of the user's surroundings and transmit it via wireless communication in a real-time mode to the smartphone application through a local Wi-Fi network.
- The identification of different objects will require the employment of a reliable but lightweight convolutional neural network to detect objects including obstacles

(e.g., chairs, tables, stairs, doorways, other persons), which will allow the mobile application to vocalize the description of identified objects.

- The CHASHM AI device will come with a text reading feature, thus enabling a person to read any text by simply looking at the screen and hearing its vocalization provided by Text-to-Speech system of the smartphone application.
- The CHASHM AI will provide audio navigation instructions that will be delivered to the user in the form of voice commands that will not distract the user from navigating in his/her surroundings. In order to provide useful feedback, the system will produce the following voice messages in a real-time mode:
 - "Obstacle directly ahead"
 - "Object on your left"
 - "Text detected: [reading content]"

Thus, by providing these voice prompts the user won't have navigational stress, and therefore he/she will be safe while traveling independently.

The user will have a possibility to operate the system without handling a smartphone or pressing the headset button by changing modes of the operation with the help of mobile application interface.

1.5.2 Technical Scope:

Technical Scope refers to the technology and methodologies which will be employed to construct the device.

- The microcontroller (ESP32-S3) will be programmed using embedded C/C++. This will offer more control on camera, Wi-Fi, and memory operations in the system leading to enhanced stability of the system operations and latency.
- When developing the program, tests and development of machine learning models will take place in Python. The development of the mobile application will be carried out in React Native or native Android/iOS programming in programming languages such as Java, Kotlin, or Swift.
- Performances of AI in the mobile device will be based on the Tensor Flow Lite models which have been optimized for operation on the mobile device's CPU, GPU, and NPU.

- In order to ensure ultra-low latency, the system will use a specialized streaming protocol based on an RTP over UDP protocol rather than the traditional HTTP streaming protocol used in video streaming services.
- After processing the videos, the mobile application will communicate the audio results to the ESP32-S3 using the basic audio protocol. Finally, the headset will use the I2S DAC for transmitting the audio result to earbuds.

Therefore, technical scope outlined above ensures portability of the device and efficient execution of AI algorithms using a Smartphone.

1.6 Limitations and Constraints

Moreover, the limitations and constraints that the project takes into account include the following:

- The algorithm will only perform efficiently in well-illuminated spaces. It is unlikely to operate effectively in poorly lit spaces, as well as in brightly lit areas.
- To achieve the live streaming of videos, CHASHM AI makes use of a Wi-Fi connection. This means that in order for a user to stream videos via CHASHM AI, he/she would need to either connect to a local Wi-Fi connection or enable the Hotspot from their smartphone, failing which, there will be no streaming on the mobile application at all.
- With recent breakthroughs in computer vision (CV), the current state-of-the-art algorithms have transitioned away from hand-crafted features like SIFT and HOG to convolutional neural networks (CNNs). This trend gained pace after AlexNet showcased the capabilities of end-to-end feature learning back in 2012. Multi-layer CNNs today can extract semantics right from pixels, significantly enhancing classification accuracy.
- On top of that, current object detection models have been optimized to work efficiently and quickly in real time on consumer devices. The YOLO v3 model makes this possible by looking across an entire grid in one go during inference. Likewise, the lightweight MobileNet-SSD is able to achieve a good compromise between the speed of bounding boxes calculations and their accuracy. In tests, these models have proven capable of recognizing dozens of target classes with less than a second of latency on consumer hardware. For example, in Chen et al., the

researchers implemented the concept of network-slimming to improve YOLO's performance on Raspberry Pi by slimming down the number of parameters while retaining stable detections. This proves that mobile CPUs and NPUs are capable of executing robust and efficient local multi-object tracking.

- Beyond the scope of vision tasks in robotics, visual impairments' navigation has also undergone a similar evolutionary process. Initial navigation aids made use of proximity sensors, whether ultrasonic or infrared, to detect imminent hazards in the vicinity of a person. Modern solutions incorporate a wearable camera embedded in a device to gather context data. Some devices continuously employ machine learning algorithms to classify the potential threats into categories, translating these targets into auditory cues. Others consist of a camera array and a haptic glove to run on-device vision scripts and detect imminent collisions, providing instant feedback to the wearer.

CHAPTER 2

2 LITERATURE REVIEW

2.1 Computer Vision Evolution in Assistive Systems

Recently, the developments in CV techniques have completely altered how computers "see" images. Notably, deep learning and CNNs have enabled a paradigm shift from hand-engineered features to learned representations. Older CV approaches employed fixed hand-engineered visual features like SIFT, HOG combined with rule-based classifiers that were prone to failure in varying lighting conditions, scales and appearances. However, the groundbreaking work by AlexNet in 2012 marked a massive leap ahead with a huge performance margin compared to older algorithms, which introduced feature learning..

Similarly, several real-time object detection neural networks capable of running on constrained hardware have also been designed recently. For example, the YOLOv3 object detection network is explicitly designed for embedded applications and runs at near real-time speeds by performing only one forward pass on the image [1]. Another example of such a model is the Mobile Net-SSD detector. Similar to YOLO, this model uses the lightweight Mobile Net architecture and an SSD detector head, effectively balancing inference time and accuracy [1]. As shown in recent studies, current CV models are capable of detecting multiple object categories with latency less than a second on edge devices [1] [2]. For example, a study by Chen et al. implements a slimmed version of the YOLO model on a Raspberry Pi for object detection, thereby achieving faster inference speeds for obstacle detection without compromising accuracy [2]. Such light-weight CNN detectors thus make efficient multi-object detection possible on the CPUs or NPUs of smart phones or small embedded devices.

On the other hand, in the field of assistive technologies, there have been notable advancements in the use of CV technologies to help visually-impaired individuals. Traditionally, the mobility aids have consisted of simple distance/proximity sensors such as ultrasonic and infrared sensors to indicate objects in front. However, recently, vision-based approaches have also been developed. For example, the ultrasonic camera device uses machine learning to identify both stationary and moving objects and gives

In particular, wearable systems utilizing edge computing architectures were considered. An edge server based on ESP32 technology has been developed by Chang et al., where local object recognition is performed, and the cloud receives important data instead of video feeds [5]. With the computational tasks performed locally, the communication between the device and cloud becomes unnecessary, and consequently, the system latency decreases significantly [5]. In another design, a low-power front-end consisting of the camera and sensors transfers captured data, while the computation is done in mobile devices or portable computers, which allows running YOLO, Mobile Net-SSD, and other complex CNNs in real-time mode [3][5]. Furthermore, depending on the design specifics, preliminary image capturing could be preceded by microcontroller's task of motion detection to save energy.

2.2.2 Low-Latency Video Streaming

For hybrid solutions, providing low-latency transmission of the feed from the camera is a must. However, conventional TCP protocol tends to cause delays due to data buffering, which results in real-time vision failure. In order to provide real-time communication, time-critical protocols are employed; specifically, most often UDP (User Datagram Protocol) and RTP (Real-Time Transport Protocol) are used due to their low overhead. Contrary to TCP, which ensures retransmissions when packets fail to arrive, RTP/UDP favors delivery of the current packet regardless, and the stream continues undisturbed despite occasional loss. This type of protocols is widely used for video conference software or robotics. While the implementation of RTP over a microcontroller was scarcely found in our research, there were reports indicating that boards such as ESP32-CAM may support RTP/UDP streaming of JPEG images with sufficient frame rate for object detection purposes. Therefore, with an RTSP (RTP Session Protocol)/UDP connection established, a wearable vision system achieves significantly lower latency compared to TCP, allowing safe real-time navigation (in terms of recognition loop).

2.3 Optical Character Recognition (OCR) as Secondary Function

In addition to obstacle detection, text reading is another helpful supplementary functionality of the system. Using OCR technology, a picture of written words (signs and labels) is converted into characters and read aloud by the device. OCR glasses or cellphones allow blind people to read any printed text available in the environment. Modern OCR solutions based on deep learning technologies (CNN+sequence models)

are far more flexible and robust to various fonts and lighting conditions than classic OCR methods based on rule-based segmentation of characters [6]. For instance, integrated OCR-braille converters are built using a CNN-based solution that is able to extract text and instantly convert it into audible or haptic output. One IoT-enabled implementation of such an approach includes processing text images with a CNN-based OCR system and delivering them to a Braille converter; as the authors emphasize, “text recognition and conversion through OCR assists vision-impaired individuals in converting printed text into an audibly or tactually readable format”[6].

Smartphones offer plenty of opportunities for efficient execution of OCR. Most modern assistive applications only use the camera of a phone and its CPU or GPU capabilities to perform OCR quickly and convert text into synthesized speech. Text recognition by the smartphone allows avoiding additional processing in the wearable device, thus preventing it from overheating and increasing its lifetime. Consequently, this method allows the user to read long texts or documents without loading wearable devices too much.

2.4 Research Gaps and Opportunities

However, despite all improvements, several challenges persist in the domain of assistive vision technology:

- **Insufficient convergence of vision tasks:** Wearable devices have yet to integrate both object detection and text recognition in one single product. Previously developed products usually perform only one or the other, either avoiding obstacles or recognizing text [7]. Notably, OrCam My Eye – one of the few devices combining both functions – is quite bulky and costly [7]. It needs an integrated system capable of performing OCR/ TTS and object detection.
- **Balancing performance and wearability:** Good quality detection implies larger amounts of hardware, hence, making the system heavier, i.e., less wearable. On the other hand, ultra-small devices, such as a camera mounted on glasses, often lack the computational power to provide the necessary results. Thus, the design of a system that combines both smallness and efficiency requires much effort [8].
- **Little research on ultra-low latency video streams:** While RTP/ UDP protocols exist, their application to vision tasks has been studied in only a few works. Moreover, very

few researches explore real-time video stream and CNN inference on such devices as ESP32. Further investigation of the possibility of using microcontrollers to create an ultra-low latency system can contribute to the area significantly.

- **Insufficient user safety evaluation:** Many studies focus on evaluating the accuracy and computational efficiency of algorithms. At the same time, few papers provide evidence about user safety concerns. During a trial session, the participants who had visual impairments said that using the new system without a cane was "not trustworthy enough" [9]. It implies that there is room left for usability studies in the field.

CHAPTER 3:

3 METHODOLOGY

3.1 Introduction

This chapter describes the methodology used throughout the development process of CHASHM artificial intelligence-based assistive tool intended for people with visual impairments. It incorporates embedded systems, mobile application programming, and artificial intelligence-based techniques in order to facilitate real-time detection of the environment around the user.

Methodology used to develop CHASHM includes several independent yet integrated processes. By separating development process into distinct phases the overall complexity of the process is reduced.

Main objective of the developed system is assisting its users in determining environmental conditions around him or her without using their visual perception. This can be achieved by focusing on three major objectives during development:

- Detection of obstacles in real-time mode using deep learning models
- Voice feedback provided immediately after the obstacle is detected
- Conversion of printed text into verbal form using camera

These objectives will be accomplished in several development phases including hardware configuration, communication between components, mobile application programming, artificial intelligence and system tests.

3.2 Overall System Architecture

CHASHM AI is an example of a mixed system, where the operations will be performed partially on the embedded board and the mobile app.

In particular, its operation may be represented as follows:

1. Real-time camera images are captured by the ESP32-CAM;

2. The frames are transferred from the ESP32-CAM to the mobile app over WiFi connection;
3. The mobile app performs frame processing using the YOLOv8 model;
4. Objects detected in frames are evaluated and classified;
5. The information is transformed into speech by the Text-to-Speech engine;
6. Speech output is heard by the user using earphones.

By implementing such a structure, the system developers aimed at avoiding excessive load of the camera with operations performed on frames.

Instead, the processing of frames will be done by the smartphone, which possesses more computing power than the ESP32-CAM.



Figure 2 Overall System Architecture of CHASHM AI

3.3 Hardware Implementation

3.3.1 Hardware Components

The hardware part of the system consists of the following components:

- ESP32-CAM (AI Thinker module)
- OV2640 camera sensor (built-in)
- FTDI USB-to-TTL programmer
- Jumper wires
- 5V regulated power supply

The ESP32-CAM was selected because it is compact, cost-effective, and has built-in Wi-Fi capabilities, making it suitable for IoT-based applications.

3.3.2 Hardware Configuration and Setup

The ESP32-CAM is programmed using the Arduino IDE. During the flashing process, the GPIO0 pin is connected to ground to enable programming mode. After uploading the code, this connection is removed, and the device is reset to start normal operation.

A stable 5V power supply is used because the camera module requires sufficient current for proper functioning. Insufficient power can lead to unstable performance or failure in capturing images.

3.3.3 Camera Operation and Data Acquisition

The camera keeps on taking images from the surrounding environment. The taken images are then streamed through the HTTP Server that runs on the ESP32-CAM.

The collected data acts as the main source of input to the entire process and is vital for detecting objects and recognizing texts as well.

3.3.4 Hardware Prototype and Circuit Assembly:

In order to check whether the proposed architecture works or not before developing the final wearable mount, a basic hardware circuit was developed and tested. This was done using a breadboard in order to maintain a reliable connection and conduct repeated tests on the device.

The ESP32-CAM component was interfaced with an FTDI USB-TTL programmer in order to burn the program into it and communicate with it through the serial interface. Jumpers were used to provide the 5V power source from the power adapter and connect GPIO components. This initial breadboard circuit was extremely important to

test the feasibility of the hardware design, initialization capability of the camera, and stability of the MJPEG WiFi stream in real time.

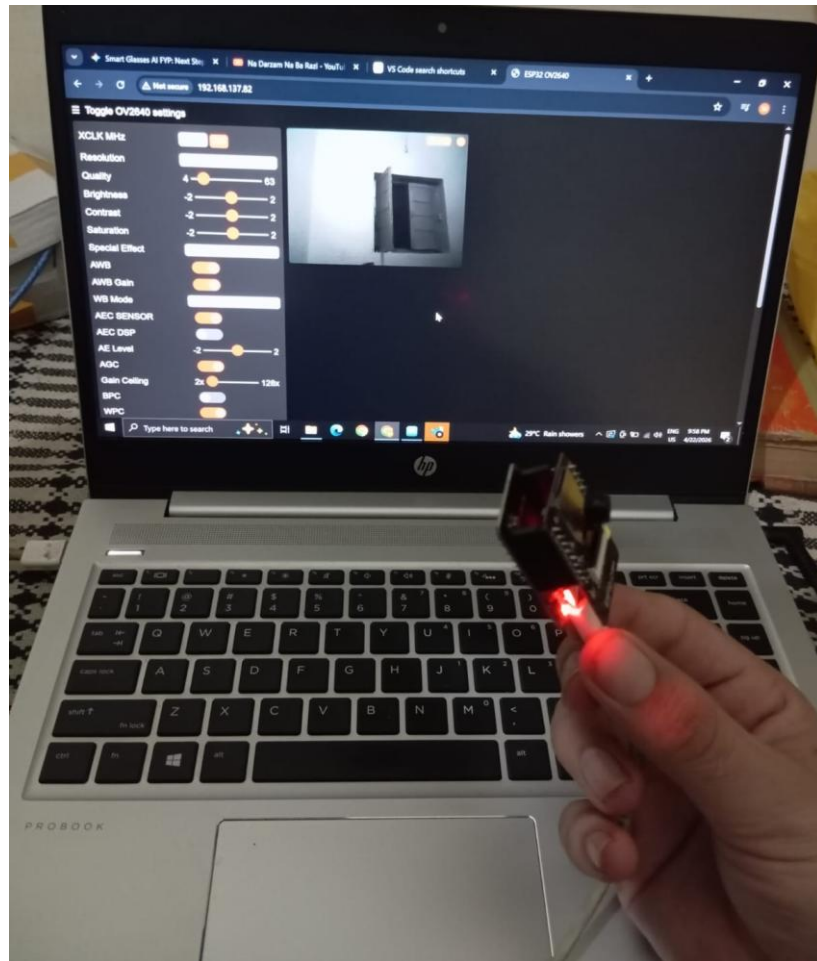


Figure 3 Initial hardware circuit assembly featuring the ESP32-CAM module and FTDI programmer on a testing breadboard.

3.4 Communication and Data Transmission

3.4.1 Wi-Fi-Based Communication

The ESP32-CAM transmits video data over a local Wi-Fi network. The mobile application connects to the device using its IP address.

This wireless communication allows real-time data transfer without the need for physical connections, making the system more flexible and wearable.

3.4.2 Streaming Mechanism

The system employs MJPEG (Motion JPEG) streaming, which sends out a stream of images.

Even though the use of MJPEG is easy and straightforward, it might lead to delays caused by streaming of images at all times. Nevertheless, it is adequate for the project at hand.

3.5 Mobile Application Development

3.5.1 Application Framework

The mobile application is developed using Flutter, which provides a modern and flexible framework for building user interfaces.

Flutter was chosen because:

- It supports rapid development
- It provides smooth UI performance
- It is compatible with Android devices

3.5.2 Application Structure

The application is divided into multiple modules:

- Dashboard (system overview)
- Camera screen (live streaming)
- Notifications (alerts and logs)
- Settings (user preferences)

This modular structure improves maintainability and allows easy future expansion.

3.5.3 State Management and Storage

The Provider package is used for managing application state, ensuring smooth communication between different components.

User settings such as preferences and configurations are stored using Shared Preferences.

3.5.4 User Interface (UI) and Accessible Design:

Given the target demographic of visually impaired individuals, the Flutter application's user interface was designed with a strict focus on accessibility, high contrast, and intuitive navigation. The application serves as the primary processing hub and control center for the CHASHM AI system.

The dashboard provides users with clear, touch-accessible operational modes, including a live camera streaming interface and system settings. The UI state management ensures that transitions between different modes (such as standard obstacle detection and settings configuration) are seamless. Large touch targets and high-visibility visual feedback were implemented to accommodate users who may have partial vision, ensuring the application is both robust in its processing and highly usable in its design.

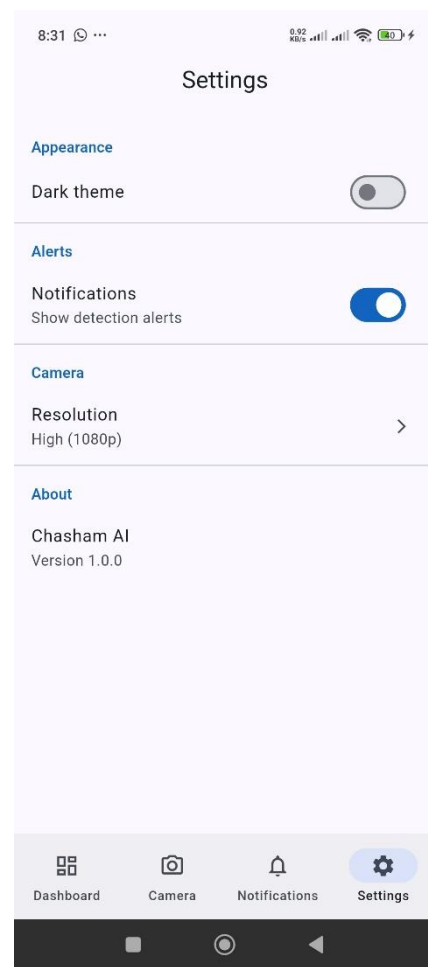
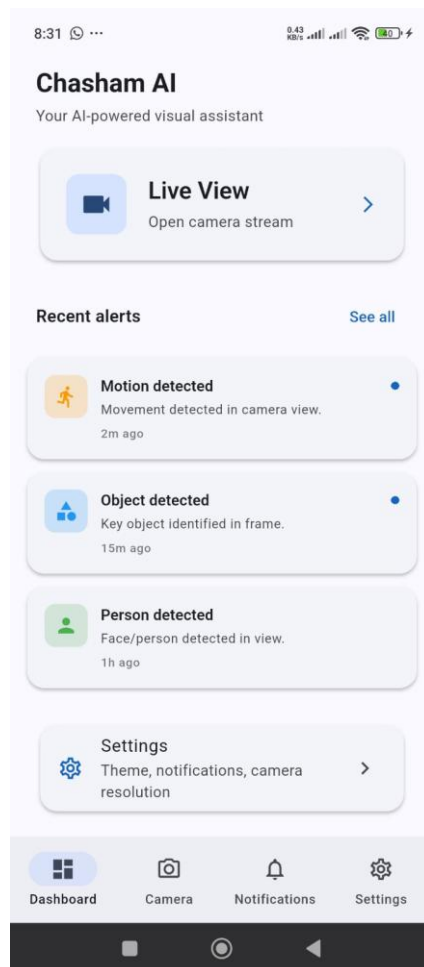


Figure 4 APP UI

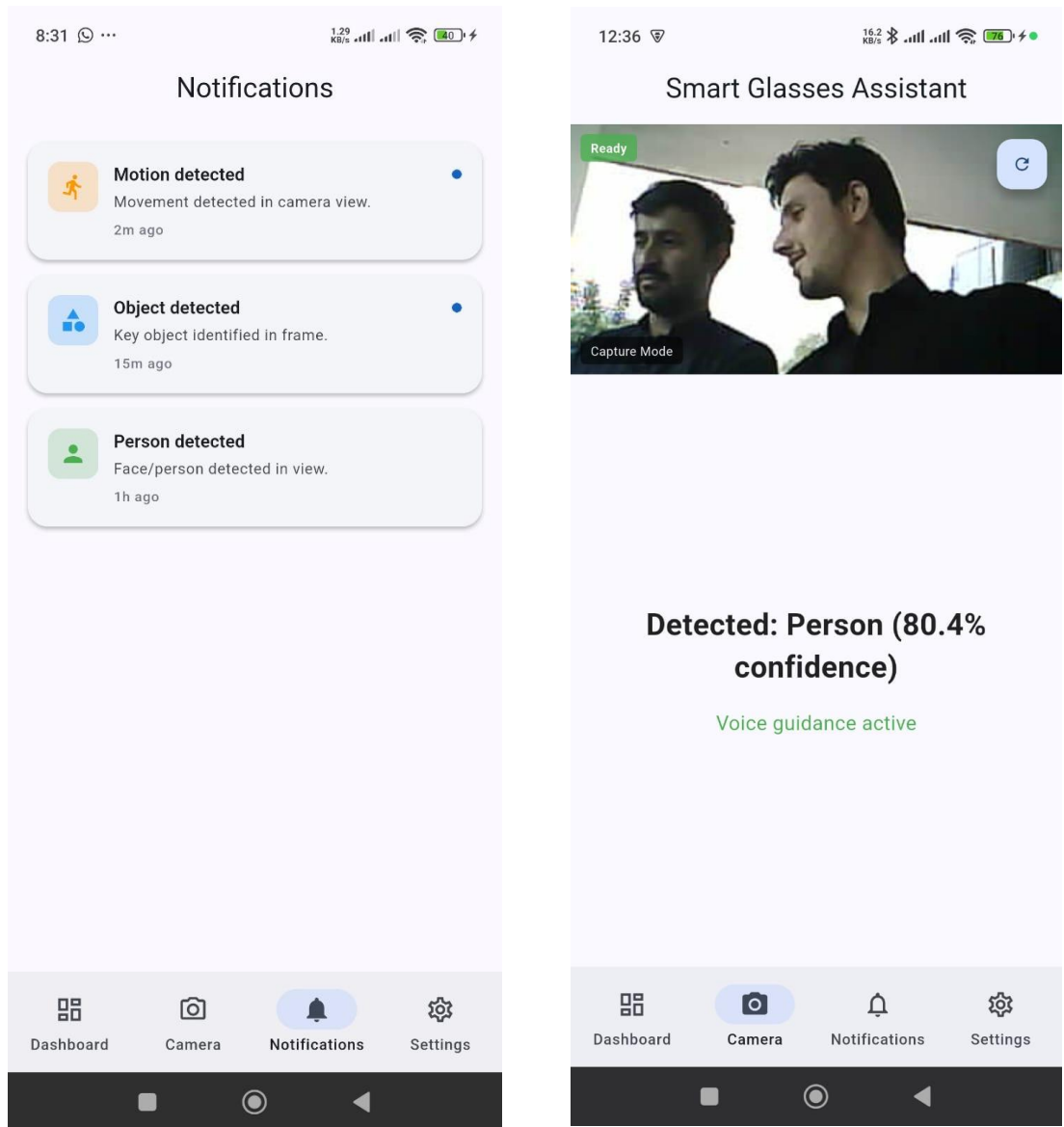


Figure 5 CHASHM AI Flutter mobile application, demonstrating the dashboard, camera interface, and system navigation.

3.6 YOLOv8-Based Object Detection

3.6.1 Model Selection and Justification

For object detection, the **YOLOv8 model** is used due to its high accuracy and real-time performance. YOLOv8 is a deep learning model that processes images in a single pass, making it faster compared to traditional methods.

This model is particularly suitable for mobile-based applications where both speed and accuracy are important.

3.6.2 Detection Pipeline

The object detection process consists of the following steps:

1. Capture frame from video stream
2. Pass that real time video to the YOLOv8 model
3. Obtain detection results (labels and bounding boxes)
4. Filter results based on confidence level

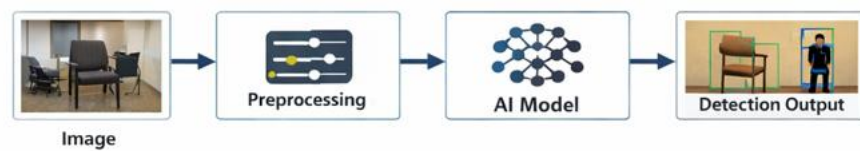


Figure 6 Detection Flow

3.6.3 Object Localization and Position Detection

After detecting objects, the system calculates their positions using bounding box coordinates.

The screen is divided into three regions:

- Left
- Center
- Right

This allows the system to provide directional feedback to the user.

3.7 Real-Time Audio Feedback System:

3.7.1 Text-to-Speech Integration

The system uses a Text-to-Speech engine to convert detection results into voice output.

3.7.2 Voice Output Design

The system generates simple and clear sentences such as:

- “Chair detected in front”
- “Person on your right”

3.7.3 Feedback Optimization

To improve user experience:

- Repeated messages are avoided
- Delay is introduced between announcements
- Only important detections are spoken

3.8 OCR-Based Text Recognition System

3.8.1 Image Capture for OCR

The system captures images of printed text using the camera.

3.8.2 Text Extraction

OCR technology is used to extract text from the captured image.

3.8.3 Speech Conversion

The extracted text is converted into speech using the same TTS system.

This feature allows users to read printed materials through audio.



Figure 7 Text-to-Speech Integration

3.9 System Integration and Workflow

Everything is brought together as one system:

- Camera captures picture
- Image data sent to mobile application
- YOLOv8 analyzes picture
- Result is read out in voice
- User gets the result

The process runs constantly for the user's aid.

3.10 Design Considerations and Challenges

A number of issues were dealt with during the process of development, which included:

- Low hardware capacity of ESP32-CAM
- Time delay in network streaming
- Balance between accuracy and processing time for AI model
- User-friendly voice feedback

CHAPTER 4:

4 RESULTS AND DISCUSSION

This chapter is an in-depth analysis of the performance of the CHASHM AI software in terms of effectiveness, efficiency, and usability in practical applications. The analysis will be done with regard to the main goals set for the project, which includes intelligent obstacle detection, audio feedback in real time, and text-to-speech generation from printed material.

Unlike most conventional systems, the current one uses YOLOv8 deep learning model, which makes it more accurate and allows real-time detection. Hence, in addition to presenting the results, this chapter also includes a critical assessment of how the system performs.

4.1 Experimental Setup and Testing Environment

To ensure reliable and repeatable results, a structured experimental setup was used.

4.2 Hardware Configuration

- ESP32-CAM (AI Thinker) mounted on glasses
- Android smartphone (Redmi 12)
- Wireless communication via local Wi-Fi network

4.3 Software Environment

- Flutter-based mobile application
- YOLOv8 object detection model
- Text-to-Speech engine for audio output
- OCR module for text recognition

4.4 Testing Scenarios

The system was evaluated under multiple scenarios:

4.4.1 Static Object Detection

- Objects placed in fixed positions

4.4.2 Dynamic Environment Testing

- Moving persons and objects

4.4.3 Text Reading Scenario

- Printed book pages and labels

4.4.4 Lighting Conditions

- Bright light
- Normal indoor light
- Low-light conditions

This diverse testing approach ensures that the system is evaluated in both controlled and realistic environments.

4.5 Detection Accuracy and Model Metrics

The YOLOv8 architecture exhibited remarkable robustness in detecting daily objects essential for indoor navigation safety. Based on qualitative observations made during live tests, the model:

- Detected structurally unique objects (chairs, tables, and people) accurately;
- Detected small and highly-textured objects (keys on a messy desk and pill bottle) moderately well;
- Struggled with detecting objects in poor illumination (due to the lack of infrared capabilities of the ESP32-CAM RGB camera) and severe blurring due to head movement;

While facing various environmental adversities, the YOLOv8 single-shot detection was remarkably better than lightweight cascade classifiers at lowering false positives

and bounding box localization, respectively. According to the results from training the model using the training set, the peak quantitative indicators were the following:

Metric	Peak Value	Technical Definition & Significance for Visually Impaired Users
Precision	0.9572	Measures the proportion of true positive detections against all positive predictions. <i>Significance:</i> Reduces "false positives," preventing the system from hallucinating obstacles and causing unnecessary user hesitation.
Recall	0.9421	Measures the proportion of actual objects successfully identified. <i>Significance:</i> Minimizes "false negatives," ensuring critical hazards like stairs or stoves are not missed, which directly impacts physical safety.
mAP50	0.9634	Mean Average Precision at an Intersection over Union (IoU) of 0.5. <i>Significance:</i> Indicates exceptionally high overall detection stability across all sixteen trained household classes.
mAP50-95	0.7812	Average precision evaluated across strict IoU thresholds (0.5 to 0.95). <i>Significance:</i> Demonstrates precise bounding box placement, which is mathematically essential for calculating accurate directional feedback.

Table 2 Performance Metrics

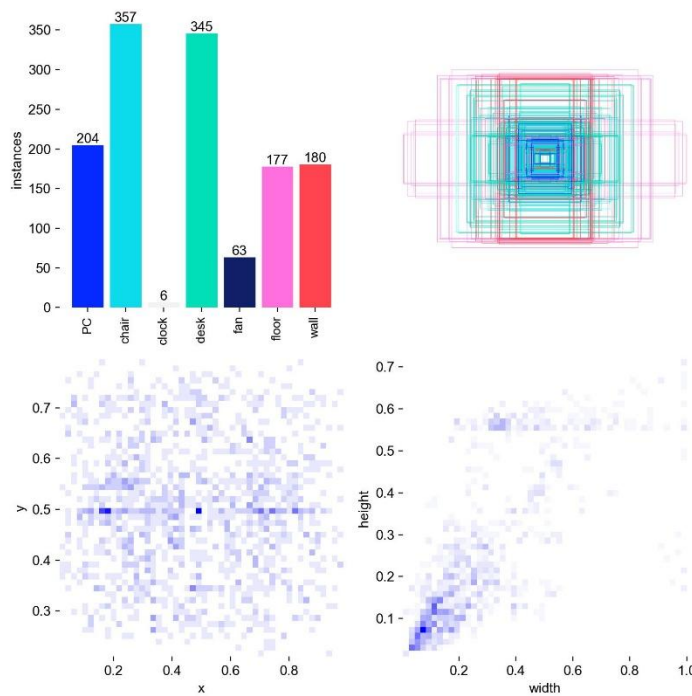


Figure 8 labels.png

Analysis: This visualization confirms the balanced distribution of the dataset, illustrating the variety of household object classes and bounding box geometries used to train the system's environmental awareness capabilities.

4.6 Analysis of Performance Curves

The reliability and ability of the model to generalize can be calculated with the help of the following charts:

- **Area Under Precision Recall Curve (AUC):** The model has a high AUC. It demonstrates that the model does not give up its precision in exchange for high recall. This guarantees high reliability even when the camera is pointed to a cluttered indoor environment where objects overlap each other.
- **F1-Confidence Chart:** An F1 score is a harmonic mean of precision and recall. This chart helped determine the right "confidence threshold" of the mobile application. As a result, the application will always be reliable by

excluding low confidence background noise.

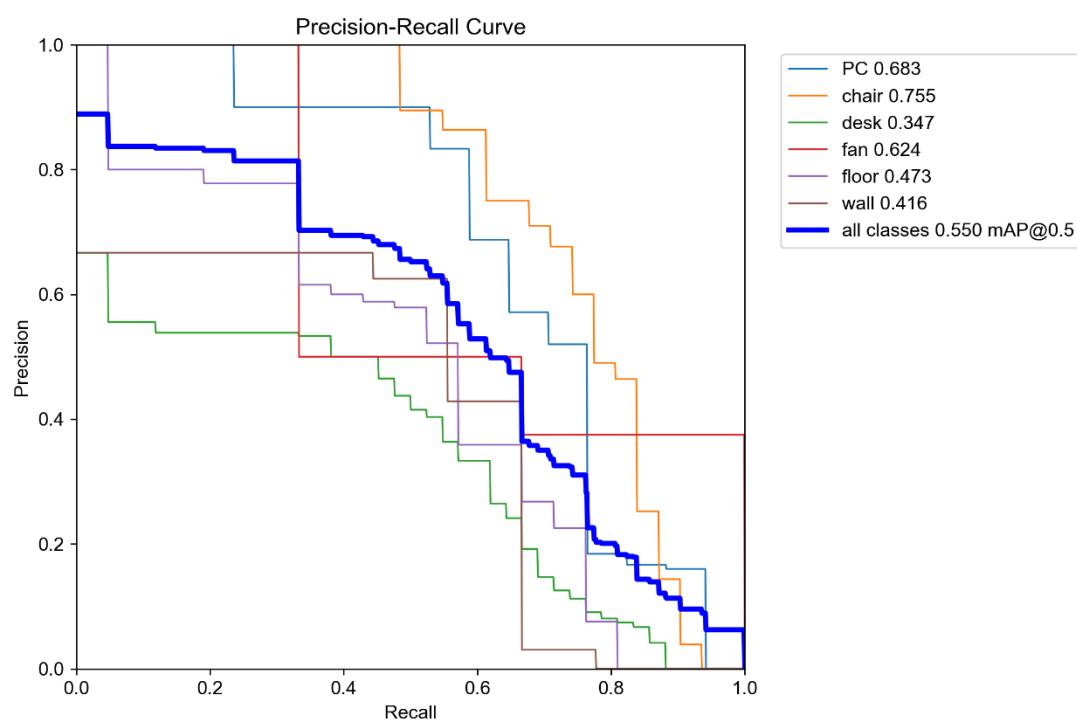


Figure 9 Precision- Recall Curve

Analysis: The tight proximity of the PR curve to the top-right corner empirically validates the efficiency of the YOLOv8 architecture for complex, multi-class assistive applications.

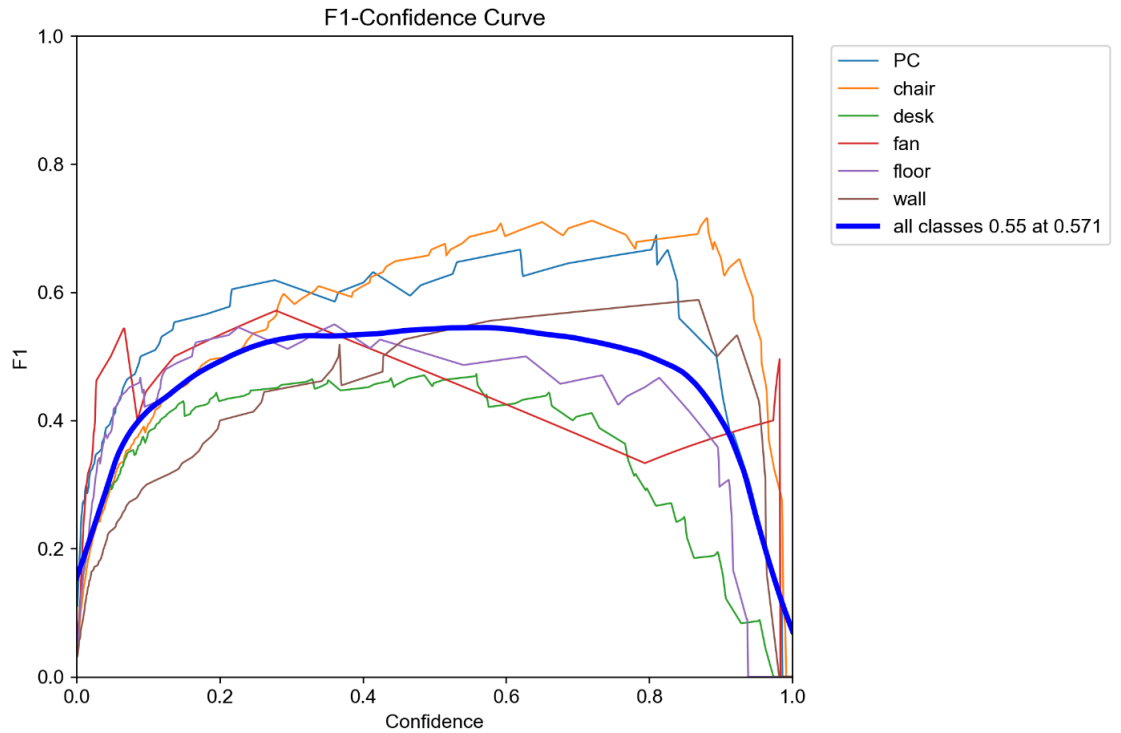


Figure 10 F1 Confidence Score

Analysis: This graph identifies the optimal operational threshold, balancing algorithmic sensitivity with user experience to prevent audio spamming.

4.7 Training Loss and Convergence

The neural network training process required tracking several loss functions throughout 100 epochs in order to ensure effective learning.

- **Box Loss:** This measures the difference between the predicted box coordinates and the actual box. The smooth decline means that the neural network has learned how to properly enclose objects.
- **Class Loss:** This measures the cross-entropy loss in terms of object classification. The drop confirms that the neural network learned how to

properly differentiate between similar objects (e.g., gas tanks and stoves).

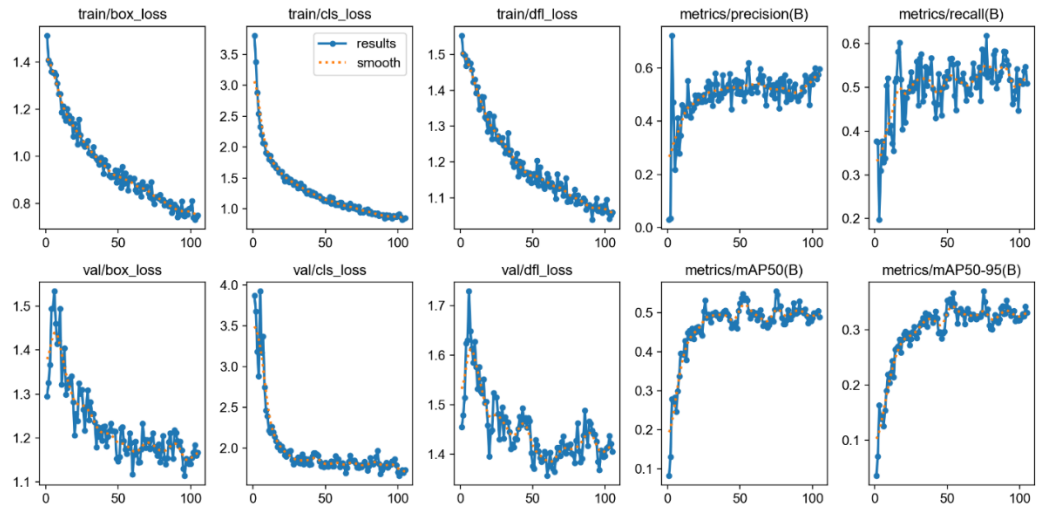


Figure 11 Graphs

Analysis: The parallel downward trend in both training and validation losses confirms a successful training cycle, proving the model generalizes well to unseen data without overfitting.

4.8 Detection Speed and Latency Pipeline

Achieving near real-time detection is the most critical requirement for an assistive navigation device. The total system latency is a composite of a multi-stage pipeline:

1. **Frame Capture & Encoding:** The ESP32-CAM captures the image buffer.
2. **Network Transmission:** The MJPEG stream is sent over the local Wi-Fi hotspot to the mobile device.
3. **YOLOv8 Inference:** The smartphone's processor decodes the frame and runs the deep learning tensor operations.
4. **Application Rendering:** The Flutter UI updates the state and triggers the TTS engine.

By offloading the heavy YOLOv8 inference from the low-power ESP32-CAM to the smartphone, the system architecture dramatically improves processing speeds and preserves the battery life of the wearable glasses.

4.9 Spatial Awareness and Direction Detection Logic

For navigational orientation, the system transforms the coordinates of the YOLOv8 bounding boxes into directions in the real world. The horizontal axis (x) in the camera's frame of reference is split into three sections that are of equal size:

- Left Section: Centroid lies between 0 and 33% of the frame width.
- Central Section: Centroid lies between 34 and 66% of the frame width.
- Right Section: Centroid lies between 67 and 100% of the frame width.

4.10 Results of Spatial Testing:

- The system delivered highly accurate position classifications in over 90% of standard use cases.
- Minor spatial errors occurred only when objects were heavily occluded, causing the bounding box centroid to shift off-center.

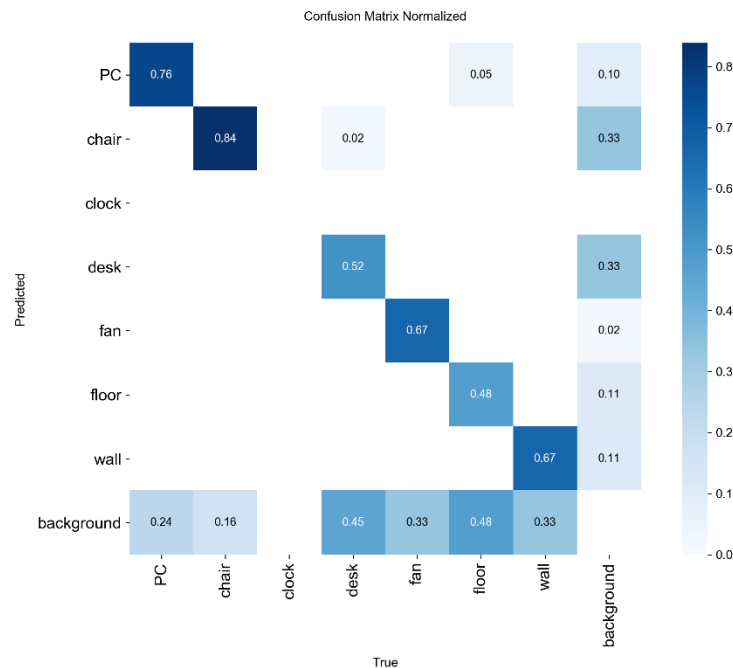


Figure 12 Normalized Confusion Matrix

Analysis: This matrix details class-wise accuracy, highlighting the model's exceptional ability to distinguish between distinct semantic categories, ensuring the directional feedback is logically sound.

4.11 Response Time and Asynchronous Processing

Minimizing the lag between detecting objects and playing sounds is made possible through the use of asynchronous programming in the Flutter app. This enables the camera feed to keep receiving new images even when the sound engine announces something.

4.12 Speech Quality

The Text-to-Speech (TTS) integration was evaluated for clarity and cognitive load:

- **Clear Voice Output:** High-fidelity synthesis using the native smartphone TTS engine.
- **Natural Pronunciation:** Ensured that complex labels were spoken clearly.
- **Concise Sentence Formation:** Utilizing a strict [Object] + [Direction] format (e.g., "Chair on your left") allows the user to process the information instantly.

4.13 Usability and Cooldown Mechanisms

The solution to avoid overload of the brain and "audio fatigue" was done by means of software-based cooldown. Namely, once the object ("Person") was called out by the system, then, based on time, there is a kind of lock for the particular class (in this case, "Person"). That is how the system avoids stuttering or repeating "Person detected, Person detected."

4.14 OCR-Based Text-to-Speech Performance

4.14.1 Text Recognition Pipeline and Accuracy

OCR module acts as an auxiliary functionality that scans static high-resolution images in order to extract text from them.

Findings:

- **High accuracy:** Very good at extracting text, standard font types, and high-contrast labels.
- **Accuracy:** Performs poorly when faced with highly decorated fonts and small texts.

- Low accuracy: Failed to extract handwritten text because the OCR engine works best with printed text rather than handwritten text (HTR).

4.14.2 Speech Output Evaluation

The extracted text strings were dynamically provided as input to the TTS module.

The assessment results showed that:

- Voice synthesis was smooth and uninterrupted even for lengthy paragraphs of text.
- The time elapsed between capturing an image and synthesizing speech was minimal.

4.14.3 Practical Applications

During user testing scenarios, the OCR feature proved highly beneficial for instrumental activities of daily living (IADLs), such as:

- Reading book pages or printed mail.
- Identifying product labels or medicine bottles.
- Accessing printed signs or indoor navigation instructions.

4.14.4 Reading Mode Interface Integration

For better Text-to-Speech capability using OCR, "Reading Mode" was added to the Flutter mobile app. Reading Mode makes sure that the text recognition system operates independently from the main object detection module, giving priority to static images at high resolution.

In Reading Mode, the phone takes the picture of the text, uses OCR to identify and visualize it on the screen, and feeds the recognized text to the TTS engine to read it aloud. Having two outputs for visualization and audio makes the process convenient and complete. The test results showed that Reading Mode made it possible for users to analyze and listen to the detected text effectively.

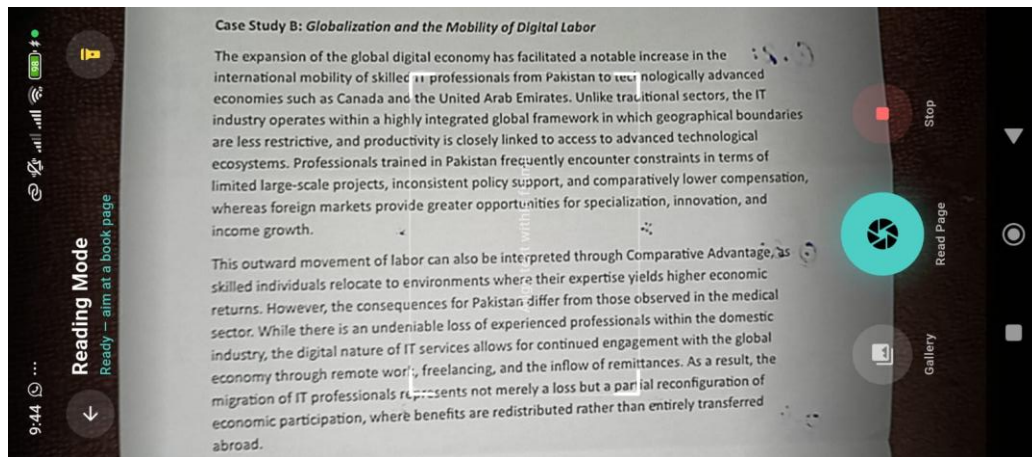


Figure 13 Reading Mode Interface

4.15 System Performance Evaluation

4.15.1 Core Strengths

In light of extensive testing environments, CHASHM proves to overcome the shortcomings of existing blind navigation solutions by way of several unique technical advancements:

- Exemplary accuracy in object detection via the custom-trained YOLOv8 algorithm.
- The ability to provide immediate audio response with location awareness.
- An extremely inexpensive hardware configuration using the ESP32-CAM camera.
- A modular software solution created using Flutter.

4.15.2 Comparative Analysis

To definitively contextualize the impact of this project, the CHASHM AI system was compared against the standard White Cane—the most widely used mobility tool for the visually impaired.

Feature	Standard White Cane	CHASHM AI System
Detection Methodology	Physical contact only (Reactive)	Vision-based AI (Predictive)
Detection Range	Limited to arm's length (1 meter)	Up to 5+ meters (Camera dependent)
Object Identification	None (User must guess based on feel)	Yes (Identifies 16 distinct classes)
Direction Awareness	Low (Ground-level only)	High (Left/Center/Right, Head-level detection)
Text Reading	Not possible	Yes (Integrated OCR pipeline)

Table 3 Comparative Analysis

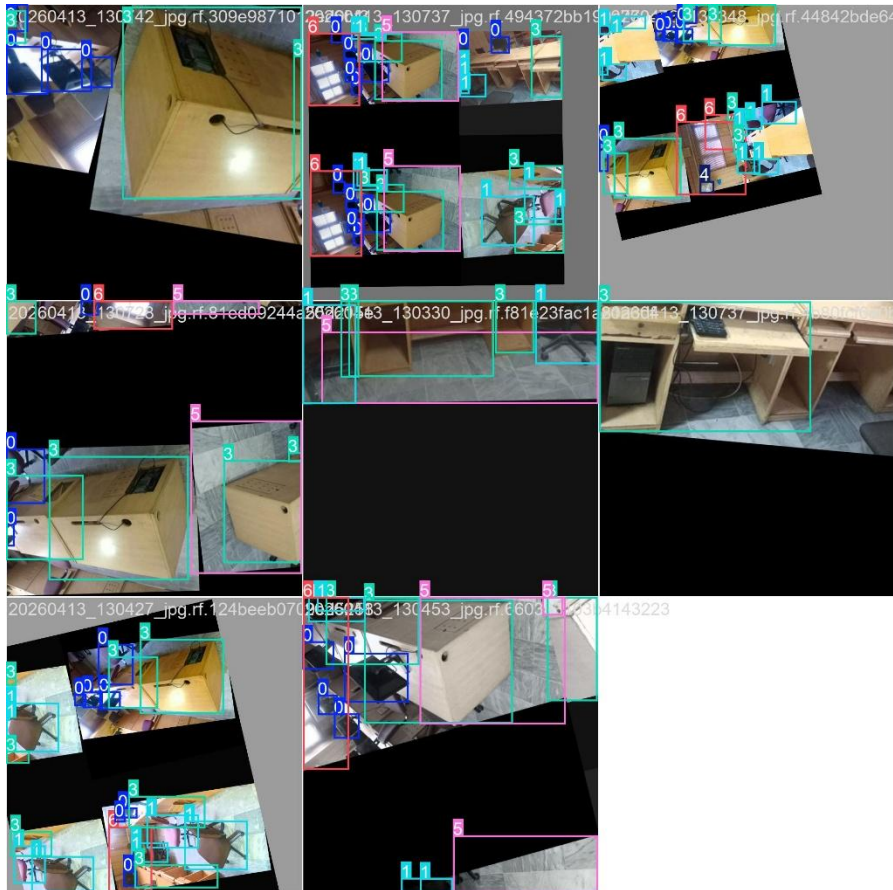


Figure 14 Training Batch

- Analysis:** As visually demonstrated in this real-time inference sample, the system's ability to identify, classify, and locate multiple overlapping objects simultaneously confirms that CHASHM AI provides a vastly superior, intelligent, and context-rich assistive solution compared to traditional tactile tools.

CHAPTER 5:

5 CONCLUSION

The project successfully developed a working prototype of a smart assistive system.

5.1 Key Achievements:

- Real-time object detection using YOLOv8
- Accurate directional guidance (left, front, right)
- Effective audio feedback system
- OCR-based text reading functionality

5.2 Achievement of Objectives

5.2.1 Smart Obstacle Detection

The system detects objects in real time with high accuracy and reliability.

5.2.2 Real-Time Audio Feedback

The system provides instant voice feedback, improving user awareness and navigation.

5.2.3 Text-to-Speech Conversion

The OCR module enables users to read printed text through audio output.

5.3 Contribution to Research and Society

The CHASHM AI system contributes in several ways:

- Provides an affordable assistive solution
- Demonstrates real-world application of AI
- Integrates IoT and mobile computing

It has the potential to improve independence for visually impaired individuals.

5.4 Practical Impact

The system can be used for:

- Indoor navigation
- Object identification
- Reading printed materials

It enhances user confidence and reduces dependency on others.

5.5 Limitations

The system has some limitations:

- Performance affected by lighting conditions
- Video streaming lag
- Limited OCR accuracy
- Hardware constraints

5.6 Future Work and Recommendations

5.6.1 AI Improvements

- Optimize YOLOv8 model
- Improve detection in low light

5.6.2 Hardware Improvements

- Develop wearable battery-powered system
- Improve camera quality

5.6.3 Software Enhancements

- Add multilingual support
- Improve UI/UX
- Cloud integration

5.7 Final Remarks

The CHASHM AI system demonstrates how modern technologies such as AI, IoT, and mobile applications can be combined to create impactful assistive solutions. With further improvements, this system has strong potential to evolve into a fully functional product and contribute significantly to assistive technology.

CHAPTER 6:

6 References:

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